

Spontaneous symmetry breaking in two-channel asymmetric exclusion processes with narrow entrances

Reproduction of Pronina & Kolomeisky, J. Phys. A 40, 2275 (2007)

Alessio

Department of Physics

August 23, 2026

Outline

- 1 Motivation
- 2 The Model
- 3 Single-lane TASEP (recap)
- 4 Mean-field theory for the two-channel model
- 5 Monte Carlo vs mean-field
- 6 Finite size and symmetry breaking
- 7 Computational methods
- 8 Conclusions & Outlook

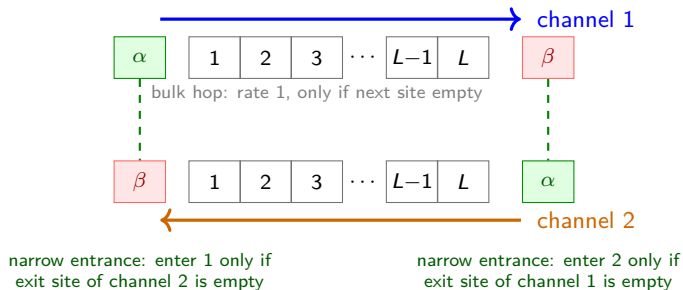
Why study exclusion processes?

- Asymmetric Simple Exclusion Process (TASEP) is the reference model of *non-equilibrium* transport — the role of the Ising model for equilibrium statistical mechanics.
- Minimal description of interacting particles on a 1D lattice with hard-core exclusion.
- Applications: motor proteins on cytoskeletal filaments (kinesin/dynein), ribosome motion in mRNA translation, car traffic, nanopore transport.

Motivation for two channels

In cells, kinesin and dynein walk *in opposite directions* on the same microtubule. Two lanes of particles moving in opposite directions, coupled at the boundaries — can they *break symmetry* spontaneously?

The paper's question



Steady states

Symmetric: $\rho_1 = \rho_2$ (both lanes equal).

Asymmetric: despite exact symmetry of the rules, one lane dense, the other dilute — $\rho_1 \neq \rho_2$.

The model: two-channel ASEP with narrow entrances

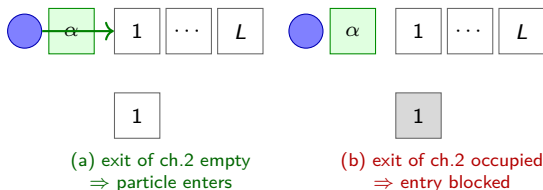


1 frame = 1 event. Bulk hop: rate 1, one site, only if next site empty. Ends: enter $\alpha=0.90$, exit $\beta=0.30$. A waiting dot = entrance blocked by the other lane's exit.

Click the frame to play the animation (lattice_animation.mp4).

The narrow-entrance rule

- Enter channel 1 (rate α) iff site 1 of channel 1 is empty **and** the exit site of channel 2 is empty.
- Enter channel 2 (rate α) iff site L of channel 2 is empty **and** the exit site of channel 1 is empty.



The coupling

Each channel's *entrance* is coupled to the other channel's *exit* — a shared bottleneck at each end.

What we want to compute

Steady-state properties as a function of the rates (α, β) :

- α = entrance rate of either channel (identical for both).
- β = exit rate of either channel (identical for both).
- Phase diagram in the (α, β) plane.
- Bulk densities ρ_1, ρ_2 and currents J_1, J_2 .
- The joint density distribution $P(\rho_1, \rho_2)$ — the diagnostic for spontaneous symmetry breaking (SSB).

Recap: single-lane TASEP

One channel, particles enter with rate α , hop right with rate 1, exit with rate β .

Mean-field approximation

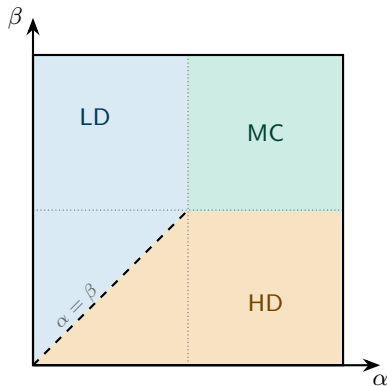
Assume no correlations between sites (eq 1 of the paper):

$$\langle p_i p_j \rangle = \langle p_i \rangle \langle p_j \rangle = p_i p_j.$$

The current across bond $i \rightarrow i+1$ is then

$$J = \langle p_i (1 - p_{i+1}) \rangle = \rho(1 - \rho), \quad \rho = \langle p_i \rangle.$$

Single-lane phases and phase diagram

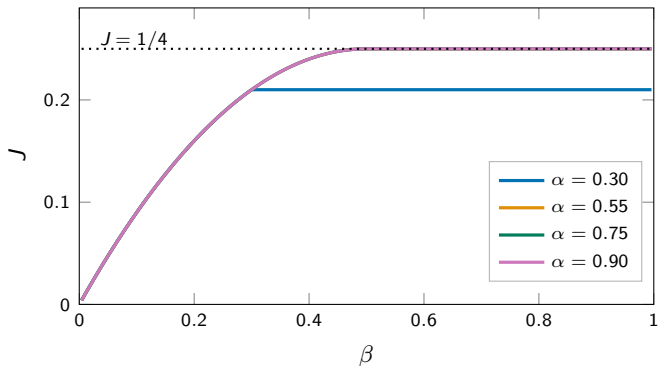


Phase	condition	J, ρ
LD	$\alpha < \beta, \alpha < 1/2$	$J=\alpha(1-\alpha), \rho=\alpha$
HD	$\beta < \alpha, \beta < 1/2$	$J=\beta(1-\beta), \rho=1-\beta$
MC	$\alpha > 1/2, \beta > 1/2$	$J=1/4, \rho=1/2$

LD = low density, HD = high density, MC = maximal current.

For a single lane, **mean-field (MFT) is exact** (matrix ansatz).

Single-lane currents



- $J = c(1 - c)$ with $c = \min(\alpha, \beta, 1/2)$: rises along $\beta(1 - \beta)$, then saturates.
- All curves with $\alpha > 1/2$ collapse onto $J = 1/4$ — the bulk becomes rate-limiting.
- For $\alpha < 1/2$ the curve stays on its own $\alpha(1 - \alpha)$ plateau — no overlap.

Strategy: two single-lane TASEPs with effective rates

Two single-lane TASEPs coupled *only through the entrances* (eq 2); the *exit* rate β is identical for both, only the entrance rates differ.

Effective entrance rates

$$\alpha_1 = \alpha \langle 1 - m_1 \rangle = \alpha(1 - m_1),$$

$$\alpha_2 = \alpha \langle 1 - p_L \rangle = \alpha(1 - p_L),$$

α_1, α_2 = entrance rates felt by channels 1, 2; m_1 = channel 2's exit density, p_L = channel 1's exit density.

Each channel is a standard TASEP with rates (α_1, β) , (α_2, β) — whose steady states we already know.

Nine candidate phases

Each channel in LD/HD/MC $\Rightarrow 3 \times 3$; filter by self-consistency of α_1, α_2 :

ch.2\ch.1	LD	HD	MC
LD	LD/LD	HD/LD	MC/LD
HD	LD/HD	HD/HD	MC/HD
MC	LD/MC	HD/MC	MC/MC

Only the **bold** combinations survive.

Symmetric phases: set-up

Notation: p_i = density at site i of channel 1, m_i = density at site i of channel 2, both counted from their own entrance ($i = 1, \dots, L$). So m_1 is channel 2's exit density, p_L is channel 1's exit density.

Symmetric ansatz (eq 6-7)

$$J_1 = J_2, \quad \rho_1 = \rho_2, \quad \alpha_1 = \alpha_2, \quad p_i = m_{L-i+1}.$$

- Both channels in the same single-lane phase.
- Symmetry + the exit current relation determine the effective rates *self-consistently*.
- Key relation (exit current of channel 2, eq 8/11):

$$J_2 = \beta m_1.$$

Symmetric MC phase (eqs 8–10)

Step 1: current fixes m_1 (eq 8). In MC, $J_1 = J_2 = 1/4$; the exit current of channel 2 gives

$$J_2 = \beta m_1 = \frac{1}{4} \quad \Rightarrow \quad m_1 = \frac{1}{4\beta}.$$

(Physical range: $\beta > 0$.) **Step 2: effective entrance rate (eq 9).**

$$\alpha_1 = \alpha(1 - m_1) = \alpha \left(1 - \frac{1}{4\beta} \right).$$

Step 3: existence condition (eq 10). MC needs $\alpha_1 > 1/2$ and $\beta > 1/2$:

$$\alpha \left(1 - \frac{1}{4\beta} \right) > \frac{1}{2} \quad \Rightarrow \quad \alpha > \frac{2\beta}{4\beta - 1}$$

Symmetric LD phase (eqs 11–13)

Step 1: current fixes m_1 (eq 11). In LD, $J_1 = J_2 = \alpha_1(1 - \alpha_1)$; exit current of channel 2:

$$J_2 = \beta m_1 = \alpha_1(1 - \alpha_1) \Rightarrow m_1 = \frac{\alpha_1(1 - \alpha_1)}{\beta}.$$

Step 2: self-consistency for α_1 (eq 2).

$$\alpha_1 = \alpha(1 - m_1) = \alpha \left(1 - \frac{\alpha_1(1 - \alpha_1)}{\beta} \right).$$

A quadratic in α_1 ; physical root (eq 12):

$$\alpha_1 = \frac{\alpha + \beta - \sqrt{(\alpha + \beta)^2 - 4\alpha^2\beta}}{2\alpha}.$$

Step 3: existence condition (eq 13). LD needs $\alpha_1 < 1/2$ and $\alpha_1 < \beta$:

$$\alpha < \frac{2\beta}{4\beta - 1}$$

Symmetric HD phase: does not exist

- In HD, $J_1 = J_2 = \beta(1 - \beta)$. From $J_2 = \beta m_1$:

$$m_1 = 1 - \beta.$$

- Effective entrance rate:

$$\alpha_1 = \alpha(1 - m_1) = \alpha\beta.$$

- HD requires $\alpha_1 > \beta$, i.e. $\alpha\beta > \beta$, i.e. $\alpha > 1$. But $\alpha \leq 1$ by definition.

Conclusion

Symmetric HD phase is impossible in the two-channel model with narrow entrances. Only MC and LD survive as symmetric phases.

Asymmetric phases: MC/HD and MC/LD don't exist

- **MC/HD** (eq 15): requires simultaneously $\beta > 1/2$ (for MC) and $\beta < 1/2$ (for HD) — contradictory.
- **MC/LD** (eqs 16–19): MC needs $\alpha_1 > 1/2$, LD needs $\alpha_2 < 1/2$. Using $J_1 = \beta p_L = 1/4$ and $J_2 = \beta m_1 = \alpha_2(1 - \alpha_2)$ leads to a condition (eq 19) that cannot hold for $0 < \alpha \leq 1$, $0 < \beta \leq 1$.

Conclusion

An asymmetric phase must pair **one channel in LD** with the other in a different phase: the candidate is HD/LD.

Asymmetric HD/LD phase (eqs 20–23)

Conditions (eq 20). Channel 1 in HD: $\alpha_1 > \beta$, $\beta < 1/2$; Channel 2 in LD: $\alpha_2 < \beta$, $\alpha_2 < 1/2$. **Step 1: currents (eq 21).**

$$J_1 = \beta p_L = \beta(1 - \beta) \Rightarrow p_L = 1 - \beta, \quad J_2 = \beta m_1 = \alpha_2(1 - \alpha_2).$$

Step 2: effective rates (eq 22).

$$\alpha_2 = \alpha(1 - p_L) = \alpha\beta, \quad \alpha_1 = \alpha(1 - m_1) = \alpha(1 - \alpha + \alpha^2\beta).$$

Step 3: existence (eq 23). HD condition $\alpha_1 > \beta$:

$$\alpha(1 - \alpha + \alpha^2\beta) > \beta \Rightarrow \beta < \frac{\alpha}{1 + \alpha + \alpha^2}$$

Asymmetric LD/LD phase: set-up (eqs 24–26)

Both channels in LD, but with *different* densities and currents.

Conditions (eq 24)

$$\alpha_1 < \beta, \alpha_1 < 1/2 \text{ and } \alpha_2 < \beta, \alpha_2 < 1/2.$$

Currents and effective rates (eqs 25–26)

$$\begin{aligned} J_1 &= \beta p_L = \alpha_1(1 - \alpha_1), & J_2 &= \beta m_1 = \alpha_2(1 - \alpha_2), \\ p_L &= \frac{\alpha_1(1 - \alpha_1)}{\beta}, & m_1 &= \frac{\alpha_2(1 - \alpha_2)}{\beta}, \\ \alpha_1 &= \alpha \left(1 - \frac{\alpha_2(1 - \alpha_2)}{\beta} \right), & \alpha_2 &= \alpha \left(1 - \frac{\alpha_1(1 - \alpha_1)}{\beta} \right). \end{aligned}$$

Two coupled quadratics in α_1, α_2 . Simplify with symmetric and antisymmetric combinations (eq 27):

$$S = \alpha_1 + \alpha_2, \quad D = \alpha_1 - \alpha_2 \neq 0.$$

Asymmetric LD/LD phase: solve (eqs 27–31)

Step 1: difference of the two equations (eq 28).

$$D = \frac{\alpha}{\beta} (\alpha_1 - \alpha_1^2 - \alpha_2 + \alpha_2^2) = \frac{\alpha}{\beta} D(1 - S).$$

Since $D \neq 0$ (asymmetric phase), we get

$$S = 1 - \frac{\beta}{\alpha} \quad (\text{eq 29})$$

Step 2: sum of the two equations (eq 30).

$$S = 2\alpha - \frac{\alpha}{\beta} \left[S - \frac{S^2 + D^2}{2} \right] \quad (\text{using } \alpha_1^2 + \alpha_2^2 = (S^2 + D^2)/2).$$

Step 3: substitute $S = 1 - \beta/\alpha$ (eq 31).

$$D^2 = 1 - 4\beta + 2\frac{\beta}{\alpha} - 3\left(\frac{\beta}{\alpha}\right)^2$$

Asymmetric LD/LD phase: effective rates (eq 32)

From S and D we reconstruct the two effective entrance rates:

$$\alpha_1 = \frac{S+D}{2}, \quad \alpha_2 = \frac{S-D}{2}.$$

$$\alpha_1 = \frac{1}{2} \left[1 - \frac{\beta}{\alpha} + \sqrt{1 - 4\beta + 2\frac{\beta}{\alpha} - 3\left(\frac{\beta}{\alpha}\right)^2} \right],$$

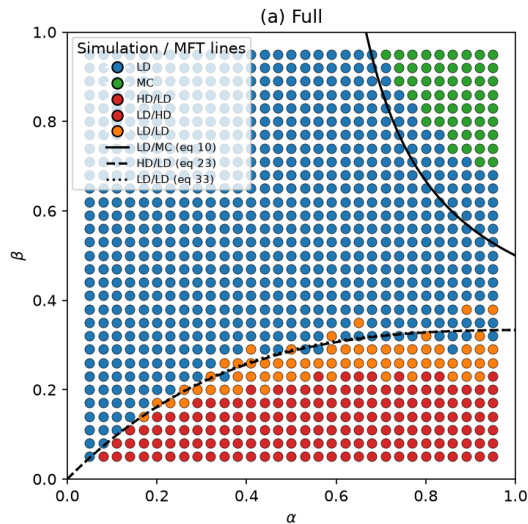
$$\alpha_2 = \frac{1}{2} \left[1 - \frac{\beta}{\alpha} - \sqrt{1 - 4\beta + 2\frac{\beta}{\alpha} - 3\left(\frac{\beta}{\alpha}\right)^2} \right].$$

Existence (eq 33)

Real roots need $D^2 > 0$; together with the LD conditions (eq 24):

$$\frac{\alpha}{1 + \alpha + \alpha^2} < \beta < \frac{\alpha(1 - 2\alpha) + 2\alpha\sqrt{\alpha^2 - \alpha + 1}}{3}$$

Mean-field phase diagram (summary)



Phase	MFT boundary
MC	$\alpha > \frac{2\beta}{4\beta - 1}$ (eq 10)
LD	$\alpha < \frac{2\beta}{4\beta - 1}$ (eq 13)
HD	never (needs $\alpha > 1$)
HD/LD	$\beta < \frac{\alpha}{1 + \alpha + \alpha^2}$ (eq 23)
LD/LD	eq 33 (narrow band)

Transitions: LD $\leftrightarrow$ MC and LD $\leftrightarrow$ LD/LD are *continuous*;
HD/LD $\leftrightarrow$ LD/LD is *first order*.

Take stock: what mean-field predicts

- ① Two symmetric phases (LD, MC) — HD impossible.
- ② Two symmetry-broken phases: HD/LD (first order) and LD/LD (continuous), in a *very narrow* band of β .
- ③ The phase diagram is fully determined by self-consistency of the effective entrance rates α_1, α_2 .

The message to verify

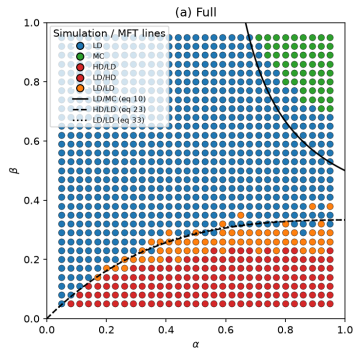
Mean-field predicts *where* transitions happen. Are these positions exact? For single-lane TASEP they are. Here, the boundary coupling may introduce correlations mean-field misses.

High-budget Monte Carlo runs

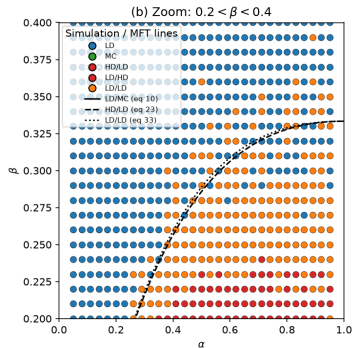
L	steps/point	steps/site
200	1×10^7	5×10^4
500	2.5×10^7	5×10^4
1000	5×10^7	5×10^4
2000	1×10^8	5×10^4

- Fenwick-tree BKL, 30 CPU workers.
- Equilibration budget must **scale with L** (steps/site $\approx cL$): a fixed budget under-equilibrates large L .
- Fig 4 scan: 2×10^8 steps/point, 40 β values ($L = 2000$).

Phase diagram: MC vs MFT ($L = 2000$)



full range

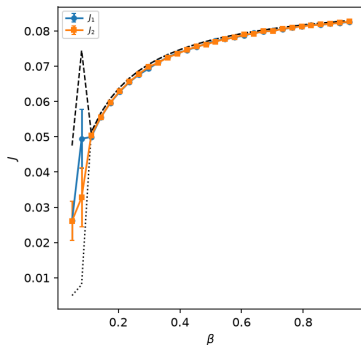


zoom: $0.2 < \beta < 0.4$

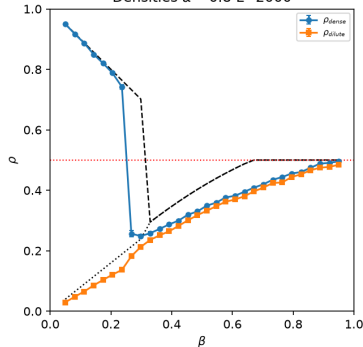
- Lines: mean-field (eqs 10, 23, 33); symbols: MC, 5×10^4 steps/site.
- Topology reproduced: LD, MC, HD/LD, LD/LD all present.
- **Boundary positions deviate** — the paper's key message.

Currents and densities ($L = 2000$)

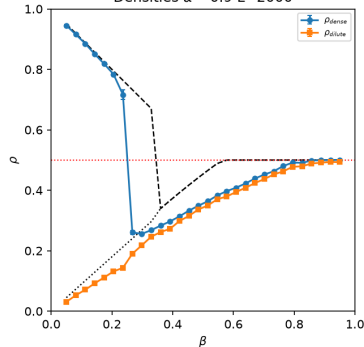
Currents $\alpha = 0.1$ $L=2000$



Densities $\alpha = 0.8$ $L=2000$



Densities $\alpha = 0.9$ $L=2000$



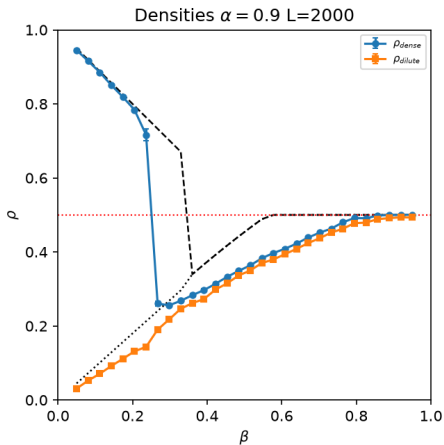
currents, $\alpha = 0.1$

densities, $\alpha = 0.8$

densities, $\alpha = 0.9$

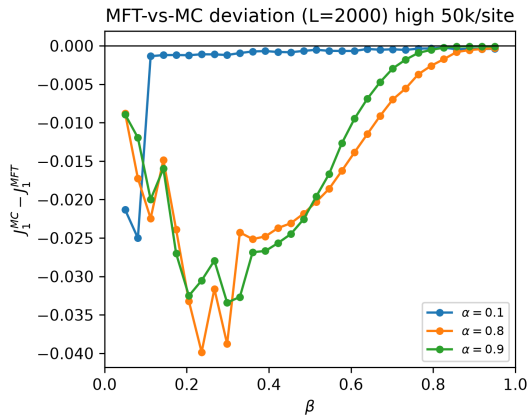
- Lines: mean-field; symbols: MC ($\pm$ s.e. over replicas).
- Good agreement in LD (low α, β); deviations grow toward the transitions.

Across the HD/LD transition ($\alpha = 0.9$)



- MC (symbols): sharp HD/LD jump at $\beta \approx 0.23$.
- MFT (dashed): jump at $\beta = 0.332$ (eq 23) — shifted by ~ 0.1 .
- Dense plateau $\rho \approx 0.71$ vs MFT ~ 0.8 .
- Both lanes $\rightarrow 1/2$ (dotted) in MC: symmetry restored.

Transition positions: the paper's key message

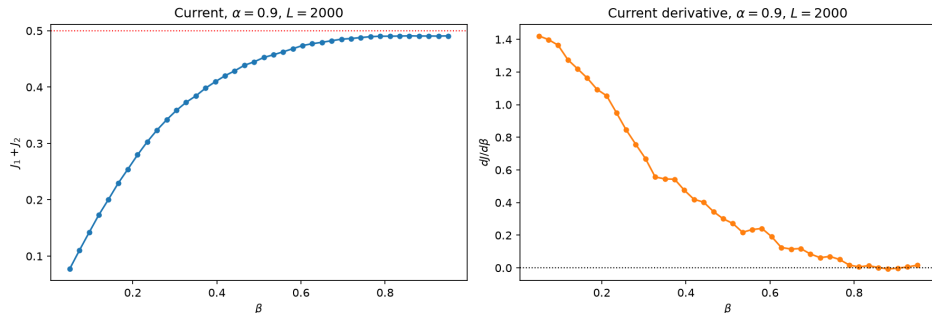


$J_1^{\text{MC}} - J_1^{\text{MFT}}$ vs β ($L = 200$)

- LD/MC boundary: MC puts it at **higher** β than MFT; deviation grows with α (+0.01 at $\alpha = 0.8 \rightarrow +0.075$ at $\alpha = 1$).
- HD/LD boundary: MC at $\beta \approx 0.25$, MFT at 0.33 (eq 23).
- Current deviation ≈ 0 deep inside a phase, **largest near transitions**.
- Unlike single-lane TASEP (MFT exact), here transition positions are only *approximate*.

Total current saturation (paper Fig 4)

Figure 4: LD/MC boundary via current saturation



- $\alpha = 0.9, L = 2000$: $J_1 + J_2$ saturates at the MC value $J = 1/2$ (dotted line).
- $dJ/d\beta \rightarrow 0$ beyond $\beta \approx 0.7$: the bulk is rate-limiting, entrance coupling irrelevant.

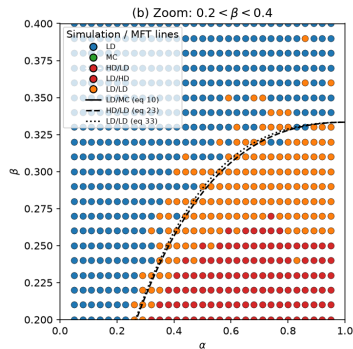
Why does mean-field fail here?

The source of the discrepancy

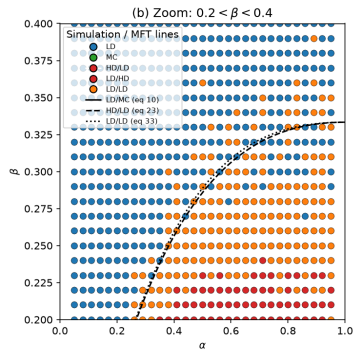
Mean-field assumes *no correlations* between sites (eq 1). The narrow-entrance coupling ties each entrance to the other channel's exit site — a strongly-correlated, local “impurity” at the boundary that mean-field cannot capture.

- These boundary correlations *propagate into the bulk* and shift the effective phase boundaries.
- The paper's own conclusion (Sec. 4): symmetry breaking in this model is a *boundary-induced* phenomenon.
- The deviation **persists up to** $L = 2000$ at 5×10^4 steps/site — looks like a genuine MFT failure, not a transient; the thermodynamic limit is still open.

The LD/LD band shrinks with L



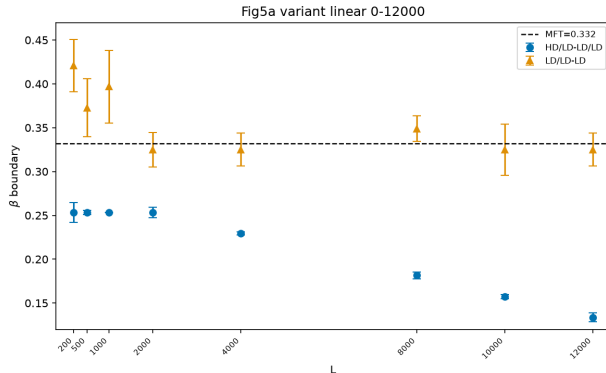
$L = 500$



$L = 2000$

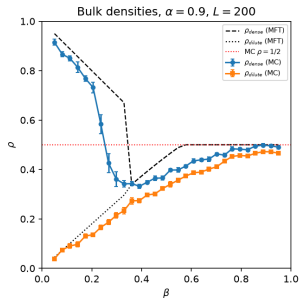
- Broken bands (LD/LD, HD/LD) **narrow** from $L = 500$ to $L = 2000$ — same budget per site.
- $L = 200$: strong, immediate HD/LD (dense ~ 0.9 , dilute ~ 0.04).
- Consistent with the paper: bands vanish in the thermodynamic limit.

Phase boundaries vs L (paper Fig 5, $\alpha = 0.9$)

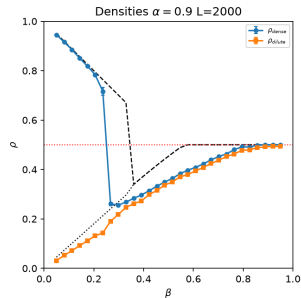


- HD/LD boundary **flat at $\beta \approx 0.253$** for $L \leq 2000$ — matches the paper (≈ 0.26), far from MFT (0.332).
- LD/LD lower boundary scatters around the MFT line.
- **Drift at $L \geq 4000$:** under-equilibration ($c = 50$; need $c \approx 100 \Rightarrow 8 \times 10^8$ steps at $L = 8000$).

Finite-size: bulk densities ($\alpha = 0.9$) vs L



$L = 200$



$L = 2000$ — 5×10^4 steps/site

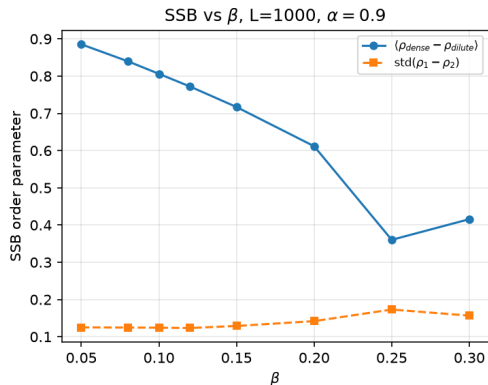
- HD/LD jump **sharpens with L** ; dilute branch unchanged.
- Position stable at $\beta \approx 0.23$ – 0.26 — **far from MFT 0.332**.
- Dense plateau ~ 0.70 ($L = 2000$) vs MFT ~ 0.80 ; error bars shrink.

SSB: a robust order parameter

Why the naive order parameter fails

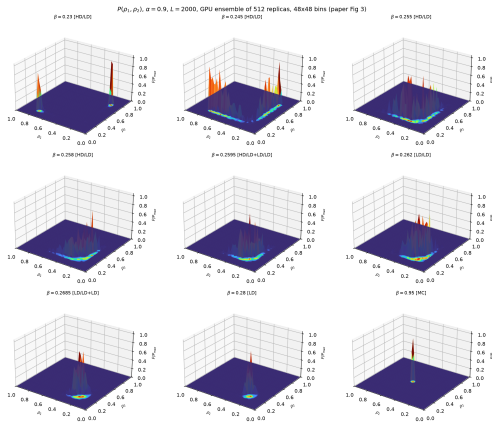
Time-averaged $|\rho_1 - \rho_2|$ is *washed out*: broken states flip between $(+, -)$ and $(-, +)$ on long time scales.

Robust choice: $\text{std}(\rho_1 - \rho_2)$ over an ensemble of replicas.



- $L = 1000$, $\alpha = 0.9$: $\langle \rho_{\text{dense}} - \rho_{\text{dilute}} \rangle$ decays $0.88 \rightarrow 0.36$ over $\beta = 0.05 \rightarrow 0.30$.
- $\text{std}(\rho_1 - \rho_2) \approx 0.12\text{--}0.17$ stays finite — flipping keeps the ensemble broad.
- SSB confined to *low* β in our MC (paper: $\beta \approx 0.23$ at the same L).

$P(\rho_1, \rho_2)$ from an ensemble (paper Fig 3)



- $\alpha = 0.9$, $L = 2000$, 512 replicas: **two peaks** (HD/LD) $\rightarrow$ arc (LD/LD) $\rightarrow$ single peak (LD) $\rightarrow$ MC center.
- Animations: [3D mp4](#) [2D mp4](#)

Simulations: BKL and the GPU ensemble

BKL (Bortz–Kalos–Lebowitz)

- Rejection-free: pick events proportional to their rates.
- Classic double-loop selection: cost $\propto n_{\text{active}}$.
- Fenwick-tree selection in $O(\log L)$: $\sim 2.4\times$ **faster, identical physics.**

Kernel	Msteps/s
classic BKL	1.8
Fenwick BKL	4.4

GPU: one thread per replica

$$\text{GPU} : \text{rep}_1 \oplus \text{rep}_2 \oplus \cdots \oplus \text{rep}_N$$

- A serial chain maps poorly to the GPU; an *ensemble* maps perfectly.
- Thousands of replicas per β — exactly what $P(\rho_1, \rho_2)$ needs.

Backend	Msteps/s
CPU, 1 thread	4.4
CPU, 30 workers	~ 100
GPU ensemble (RTX 4070)	120–180

One run $\rightarrow$ all figures

- Per- L result bundles: `results/<L>/` with `fig2`, `fig4`, `fig6`, `fig3`, `ssb` data as `.npz` + `params.json`.
- Plot scripts *read* the saved data — no MC at plot time; figures regenerate in seconds.
- Reproducible: fixed seeds, checkpointed chunks (crash-safe), incremental saves per unit.

Cost model (BKL vs Fenwick vs GPU)

Classic BKL $\propto n_{\text{active}}$ (fast at low density), Fenwick/GPU $\propto \log L$ (wins at high density). The paths cross in β .

Summary

- Reproduced the two-channel ASEP with narrow entrances: MFT phase diagram, currents, densities, $P(\rho_1, \rho_2)$ — up to $L = 2000$ at 5×10^4 steps/site.
- MFT mechanism: self-consistent effective entrance rates $\Rightarrow$ symmetric LD, MC + broken HD/LD, LD/LD.
- Confirmed the paper's message: **MFT transition positions are not exact** — HD/LD at $\beta \approx 0.25$ vs MFT 0.33; LD/MC shifted up, deviation growing with α .
- Finite size: broken bands shrink with L ; HD/LD boundary flat at 0.253 for $L \leq 2000$.
- SSB characterized robustly: ensemble $\text{std}(\rho_1 - \rho_2)$ and bimodal $P(\rho_1, \rho_2)$ (512 replicas).
- Fenwick BKL + GPU ensemble (RTX 4070, ~ 150 Msteps/s) make the scans affordable.

References



E. Pronina, A. B. Kolomeisky, *Two-channel asymmetric exclusion processes with narrow entrances*, J. Phys. A: Math. Gen. **40**, 2275 (2007).



C. T. MacDonald, J. H. Gibbs, A. C. Pipkin, *Kinetics of biopolymerization on nucleic acid templates*, Biopolymers **6**, 1 (1968).



B. Derrida, E. Domany, D. Mukamel, *An exact solution of a one-dimensional asymmetric exclusion model with open boundaries*, J. Stat. Phys. **69**, 667 (1992).



A. B. Bortz, M. H. Kalos, J. L. Lebowitz, *A new algorithm for Monte Carlo simulation of Ising spin systems*, J. Comput. Phys. **17**, 10 (1975).

Open questions

- Do the broken bands collapse onto the first-order line as $L \rightarrow \infty$? Our L -scan is consistent with it; large L needs a budget $\propto L^2$.
- Does the MFT-vs-MC boundary shift persist in the thermodynamic limit? It persists up to $L = 2000$ so far.
- Beyond MFT: two-site correlations / boundary-correlation parameters — can they close the gap?
- Extensions: asymmetric rates ($\alpha_1 \neq \alpha_2$), wider entrances, 3 channels.

Thank you!

Questions?

Anticipated questions — backups follow.

Backup: equilibration must scale with L

- A fixed steps/site budget *under-equilibrates* large L : with 3×10^3 /site, $L = 500$ gives dense ~ 0.75 vs $L = 200$'s ~ 0.92 — a systematic bias, not noise.
- Calibration probe: the SSB dense basin at $L = 1000$ needs $\sim 10^5$ steps/site to saturate (dense ≈ 0.89).
- Rule: steps/site $\approx cL$ with $c \approx 100 \Rightarrow$ total $\approx cL^2$.
- Today's runs use $c = 50$ (5×10^4 /site); the fig-5 drift at $L \geq 4000$ is exactly this budget running out.

Backup: kernel performance details

- Classic BKL: cost $\propto n_{\text{active}}$ — cheap at low β (few events), expensive at high density.
- Fenwick: $O(\log L)$ always — wins at high β ; the two cost curves cross in β .
- Per-thread GPU: $\sim 35\text{k}$ step/s (serial chain, latency-bound); $\sim 58\text{--}60\text{k}$ at $L = 8000$.

Backend	Mstep/s	rel.
CPU 1 $\times$ classic	1.8	1 $\times$
CPU 1 $\times$ Fenwick	4.4	2.4 $\times$
CPU 30 workers	~ 100	55 $\times$
GPU (RTX 4070)	120–180	$\sim 90\times$

GPU best for: phase diagrams, $P(\rho_1, \rho_2)$ ensembles, big statistics. Long single trajectories: CPU ($\sim 100\times$ faster per-thread).